

RE-ENTERING UNFINISHED MUSIC: SESSION-CONDITIONED RETRIEVAL OF PERSONAL AUDIO FRAGMENTS WITH REVERSIBLE DAW ACTIONS

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ABSTRACT

Musicians accumulate large libraries of unfinished audio fragments yet get little support deciding what to try next when re-entering a project. We formulate *session-conditioned continuation retrieval*: given the live state of a DAW project, rank a musician’s own fragments by their utility for the next creative step. Pocket Producer implements this in Audiotool: a transparent fingerprint of the live project, ranking with frozen audio–text representations, and reversible single-transaction insertion. On 1,755 weak pairs from the Freesound Loop Dataset, frozen CLAP cosine reaches 0.883 pairwise accuracy and learned rerankers (up to 335K parameters) cannot beat it; a five-parameter fusion of cosine, tempo, key, and tag signals reaches 0.794 on the hardest negatives, +4.0 pp over cosine (0.754; one-sided sign test $p = 0.031$). Session-structure signals (instrument role gap, harmonic compatibility) show no gain on pack-membership labels, while an exploratory alignment of the same signals with external rater preferences (9 raters, 20 pairs, Fleiss’ $\kappa = 0.294$) points the opposite way—weak-label and human evaluations appear to reward different structure, motivating task-specific evaluation.

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the central question is not “what sounds similar?” but “what would be useful as the next step?” Existing work on loop compatibility [1–3] and stem coherence [4, 5] asks whether two elements can co-exist in a mix; our query is a *live session state*, and the candidates are the musician’s own creative memory, not a public catalog.

We define **continuation utility** as the property that a candidate fragment helps the musician make progress in the current session—subsuming tempo fit, timbral contrast, structural gap-filling, and stated intent. We contribute:

1. **Task**: session-conditioned ranking of personal fragments with a preregistered primary metric (pairwise



Figure 1. Pocket Producer retrieval loop: live session fingerprint $\rightarrow$ ranking $\rightarrow$ reversible insertion $\rightarrow$ preference feedback.

accuracy over human preferences).

2. **System**: a working prototype in a real DAW (Audiotool) with live document reading, reversible insertion, and explainable evidence.
3. **Protocol**: a backbone bake-off, a baseline comparison (learned rerankers lose to frozen cosine; multi-signal fusion gains +4.0 pp on hard negatives), and a blind A/B evaluation with 9 external raters.

2. POCKET PRODUCER

Figure 1 shows the loop; the system reads and writes Audiotool projects through the Nexus SDK.

2.1 Session Fingerprint

The live Nexus document is mapped deterministically to a `SessionFingerprint` (tempo, track counts and types, playhead, user-editable intent); only this summary leaves the browser, and missing fields are never imputed. Session audio regions are encoded with MS-CLAP 2023 [6] into a mean session embedding (CLAP’s joint audio–text space also serves text intent); per-region instrument roles and 12-dim CQT chroma are extracted for the session-structure signals below.

2.2 Continuation Utility Ranking

Candidate fragments are scored in two stages. First, per-space retrieval runs independently over four signal spaces: (1) structured metadata (tempo/key/tag matching), (2) frozen CLAP embeddings (cosine similarity), (3) *role-gap fill*—a zero-shot instrument probe boosts candidates whose predicted role fills a gap in the session, and (4) *harmonic compatibility*—transpose-invariant cross-correlation of CQT chroma vectors scores tonal fit



Table 1. Backbone bake-off on 2,558 FSLD items. R@10 = same-source Recall@10; Genre/Role = linear-probe accuracy.

Backbone	Dims	R@10	Genre	Role	Bytes/frag
laion-clap-music	512	0.713	0.415	0.556	2,048
msclap-2023	1024	0.692	0.384	0.533	4,096

Table 2. Baseline comparison on FSLD weak pairs (LAION-CLAP 512-dim, 5 seeds). Hard-sim = hard_similar negatives; Hard-tk = hard_tempo_key. Bold = best per column.

Model	Params	Overall	Hard-sim	Hard-tk
cosine-mean	0	0.883	0.754	0.911
deepsets-v1	164K	0.735	—	—
pocketrank-v1	335K	0.742	—	—
cosine+session	2	0.882	0.749	0.911
fusion (5-signal)	5	0.894	0.794	0.897
fusion (7-signal)	7	0.892	0.792	0.894

across all 12 semitone shifts. Results are fused with Reciprocal Rank Fusion [7]; a learned reranker can be applied on top, and a rule baseline with structured evidence codes is the always-available fallback.

2.3 Reversible Audiotool Interaction

A recommendation can be previewed and inserted in a single Nexus transaction; an `InsertReceipt` records created entity IDs so undo restores the exact prior state (verified with the official WASM validator); interactions are logged as preference feedback.

3. PRELIMINARY EVALUATION

3.1 Backbone Bake-off

On 2,558 items from the Freesound Loop Dataset (FSLD) [8], LAION-CLAP Music [9] (512-dim) outperforms MS-CLAP 2023 (1024-dim) on all three probes (Table 1) at half the storage. We deploy MS-CLAP 2023 because its training-data license audit is complete—quality and eligibility are separate decisions.

3.2 Baseline Comparison and Multi-Signal Fusion

We constructed 1,755 leave-one-out pairs from FSLD, using same-pack membership as a weak proxy for continuation utility, with three negative types: *easy* (tempo or key incompatible), *hard_tempo_key* (tempo and key both compatible), and *hard_similar* (highest embedding cosine from another pack). Splits are grouped by source pack; Table 2 reports 5-seed mean validation pairwise accuracy.

Frozen cosine similarity (0.883) is already a strong baseline: two learned rerankers with up to 335K parameters trained with BPR loss [10] score below it (0.735, 0.742), overfitting to noise in the weak labels. However, a five-parameter logistic fusion of audio cosine (mean and max pooling), tempo match, key match, and tag Jaccard overlap—trained with the same BPR loss—reaches **0.794**

on *hard_similar* negatives, a +4.0 pp gain over cosine alone (0.754); the gain holds in all 5 seeds (one-sided sign test $p = 0.031$). *Hard_similar* negatives are by construction cosine’s failure mode (mined as highest-cosine from another pack): the +4.0 pp measures what complementary signals recover there—the overall gain is +1.1 pp, holding in all 5 paired splits. On *hard_tempo_key* negatives, where tempo and key features are neutral by construction, fusion (0.897) stays close to cosine (0.911), ruling out label leakage.

Adding the session-structure signals (Section 2.2) does not improve the fusion: 7-signal results (0.892 overall, 0.792 *hard_similar*) are indistinguishable from 5-signal, and *cosine+session* (0.749) sits slightly below cosine alone—pack membership reflects recording context, not whether a candidate fills a compositional gap.

3.3 Author Pilot Evaluation

Under a preregistered protocol (frozen 2026-07-30), the author labeled 60 blind A/B pairs; 42 received a preference, 18 “neither” (30%). No ranker exceeds 0.571 pairwise accuracy (chance = 0.5), and cosine-agreement is 57.1% (24/42)—statistically indistinguishable from chance (Wilson 95% CI [0.42, 0.71]). The role-gap signal fires on only 7 of 42 pairs but agrees in 5 of 7. Author labels are excluded per the preregistration.

3.4 External Blind Evaluation

Nine external raters labeled the same 20 blind A/B pairs (180 labels). Fleiss’ $\kappa = 0.294$ indicates fair inter-rater agreement—typical for subjective music-preference tasks [11]—with 10 of 20 pairs reaching $\geq 7/9$ majority. Most pairs elicit a clear preference (only 8.9% “neither” versus the pilot’s 30%), suggesting context information aids decision-making.

In a post-hoc exploratory analysis (not preregistered; five signals compared), we align each signal with the per-pair human majority. On rater-consensus pairs ($\geq 7/9$), harmonic compatibility matches the majority in 7 of 9 cases, CLAP cosine in 6, max-pooled cosine in 4, and tempo—the strongest theory signal on weak labels—in 3; the sparse role-gap signal is correct on all 3 pairs where it fires, echoing the pilot’s 5/7. No single difference is significant at $n = 9$, but the direction is consistent: the signals that win on weak labels (Table 2) lose ground on human preferences and vice versa, suggesting the two evaluations reward different structure and motivating a preregistered replication.

4. DISCUSSION AND LIMITATIONS

Weak labels $\neq$ ground truth. Weak labels compare signals at scale but cannot establish continuation utility—and can invert the human signal ranking (Section 3.4). Offline numbers use LAION-CLAP; fusion on deployed MS-CLAP is untested.

Scope and privacy. We measure perceived relevance, not creative quality; session audio is never stored and OAuth tokens never leave the browser.

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 161 who has verified each element against the linked artifacts 212
 162 and source publications. 213

163 5. REFERENCES 214

- 164 [1] B.-Y. Chen, J. B. L. Smith, and Y.-H. Yang, “Neural 216
 165 loop combiner: Neural network models for assessing 217
 166 the compatibility of loops,” in *Proceedings of the 21st*
 167 *International Society for Music Information Retrieval*
 168 *Conference (ISMIR)*, 2020.
- 169 [2] S. Lattner, “SampleMatch: Drum sample retrieval by
 170 musical context,” in *Proceedings of the 23rd Interna-*
 171 *tional Society for Music Information Retrieval Confer-*
 172 *ence (ISMIR)*, 2022.
- 173 [3] X. Yi and M. Barthet, “A generative framework for
 174 composition-aware loop recommendation in music
 175 production: Drum2Bass use case,” in *Proceedings of*
 176 *the Sound and Music Computing Conference (SMC)*,
 177 2024.
- 178 [4] R. Ciranni, E. Postolache, G. Mariani, M. Mancusi,
 179 L. Cosmo, and E. Rodolà, “COCOLA: Coherence-
 180 oriented contrastive learning of musical audio repre-
 181 sentations,” *arXiv preprint arXiv:2404.16969*, 2024.
- 182 [5] J. Nistal, M. Pasini, C. Aouameur, M. Grachten, and
 183 S. Lattner, “Diff-A-Riff: Musical accompaniment co-
 184 creation via latent diffusion models,” in *Proceedings of*
 185 *the 25th International Society for Music Information*
 186 *Retrieval Conference (ISMIR)*, 2024.
- 187 [6] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang,
 188 “CLAP: Learning audio concepts from natural lan-
 189 guage supervision,” in *IEEE International Conference*
 190 *on Acoustics, Speech and Signal Processing (ICASSP)*,
 191 2023.
- 192 [7] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, “Re-
 193 ciprocal rank fusion outperforms Condorcet and indi-
 194 vidual rank learning methods,” in *Proceedings of the*
 195 *32nd Annual International ACM SIGIR Conference*,
 196 2009.
- 197 [8] A. Ramires, F. Font, D. Bogdanov, J. B. L. Smith, Y.-H.
 198 Yang, J. Ching, B.-Y. Chen, Y.-K. Wu, W.-H. Hsu, and
 199 X. Serra, “The freesound loop dataset and annotation
 200 tool,” in *Proceedings of the 21st International Society*
 201 *for Music Information Retrieval Conference (ISMIR)*,
 202 2020.
- 203 [9] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick,
 204 and S. Dubnov, “Large-scale contrastive language-
 205 audio pretraining with feature fusion and keyword-

to-caption augmentation,” in *IEEE International Con-*
ference on Acoustics, Speech and Signal Processing
(ICASSP), 2023.

- [10] S. Rendle, C. Freudenthaler, Z. Gantner, and
 L. Schmidt-Thieme, “BPR: Bayesian personalized
 ranking from implicit feedback,” in *Proceedings of the*
25th Conference on Uncertainty in Artificial Intelli-
gence (UAI), 2009.

- [11] M. Schedl, A. Flexer, and J. Urbano, “The neglected
 user in music information retrieval research,” *Journal*
of Intelligent Information Systems, vol. 41, no. 3, pp.
 523–539, 2013.